**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 194 · Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Write −2 in 8-bit two's complement.

Q2. What is unsigned 00100000 after a logical right shift by one?

Q3. Convert binary $0.1_2$ to decimal.

Q4. Calculate $0.5 \times 2^3$.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

range — Spread of magnitudes that can be represented.

## New learning

Why is 0.0110 not normalised? — For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

Shift and compensate — Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

Verify the value — $0.0110_2$ is 0.375; 0.375 × 16 = 6. $0.1100_2$ is 0.75; 0.75 × 8 = 6.

## Worked example

Normalise positive $0.0011_2 \times 2^5$.

1. First shift — 0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.

2. Second shift — 0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.

3. Check — Both represent 6 in decimal, so normalisation preserved the value.

# We Do · Practise with guidance

Normalise positive $0.0101_2 \times 2^4$.

```
0.0101 × 2⁴
→ ? × 2?
```

Predict the shift

Shift the mantissa left one position: 0.0101 → 0.1010.

Adjust and check

Exponent decreases from 4 to 3. Both representations equal decimal 5.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

### Question 2

Normalise positive $0.0010_2 \times 2^5$.

### Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

# Exit Ticket · Answer before checking

Shifting $0.0110 \times 2^4$ left to $0.1100 \times 2^4$ preserves its value.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. 11111110.

Q2. 00010000.

Q3. 0.5.

Q4. 4.

### You Do pages · Independent questions

Q1. Mantissa $0.1100_2$; exponent 3.

Q2. $0.0010 \times 2^5 \rightarrow 0.0100 \times 2^4 \rightarrow 0.1000 \times 2^3$.

Q3. Redundant leading sign bits are removed so available mantissa positions hold more significant information. Exponent adjustment preserves the represented value.

### Exit Ticket · Model answer

False. To preserve the value, the exponent must decrease to 3.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**